IBM Z DATATHON 2026 · REAL-TIME AI FOR CRITICAL DECISIONS

ZeroLag

Real-time detection and interruption of UPI payment scams, running inside the transaction on IBM LinuxONE.

SCAMS INTERCEPTED

95.9%

of fraudulent payments, held-out week

FRAUD VALUE STOPPED

98.3%

₹91.3 lakh of ₹92.9 lakh

GENUINE PAYMENTS AFFECTED

0.37%

0.05% held

DECISION TIME

4.2 ms

median, measured on IBM LinuxONE

Bengaluru hub · 17–18 October 2026

github.com/hsbofficial1/zerolag

PROBLEM

The fastest-growing UPI frauds target the payer, not the system. Each one leaves a signal that is visible at the moment of payment.

SCAM TYPE	HOW IT WORKS	SIGNAL AVAILABLE DURING THE PAYMENT
Digital arrest	Impersonators posing as police, CBI or customs keep the victim on a call and direct transfers to a "verification" account.	Payer on an active call; first payment to the receiver; amount far above the payer's norm.
Collect-request fraud	The victim is told to approve a request to "receive" a prize or refund. Approving a COLLECT request debits the account.	COLLECT from an unknown receiver; recently opened receiving account.
Account takeover	After a SIM swap or remote-access install, the attacker drains the account from a new device.	Unrecognised device and location; burst of payments to new receivers.
Money mules	Purchased or rented accounts collect from many victims and forward funds within minutes.	Many distinct payers to one receiver within a day; rapid onward transfers.

UPI processes more than 10 billion payments a month. Once a payment settles, recovery depends on slow freeze-and-refund processes. **Prevention has to happen inside the authorisation window.**

Sources: NPCI UPI product statistics; RBI and Indian Cyber Crime Coordination Centre (I4C) public advisories on digital-arrest and collect-request fraud.

DESIGN REQUIREMENTS

An effective control must decide within the payment, explain its decision and never expose customer identity.

Inline and fast

The score must return inside the authorisation call, in milliseconds, for every payment. Batch or post-settlement review cannot stop a transfer that has already settled.

Explainable

A customer who is warned needs a reason specific to their situation. An analyst who releases a held payment needs the evidence. A regulator needs an audit trail.

Private by design

Behavioural profiling must not create a new store of identifiable payment data. Identifiers are tokenised before any processing, and demographics are excluded from the model.

ZeroLag meets all three requirements in a single service on IBM LinuxONE, next to the systems of record that already process the payment.

SOLUTION

Each payment receives one of three graded decisions with plain-language reasons, so friction is applied only where risk is real.

DECISION	ACTION	SHARE OF GENUINE PAYMENTS
ALLOW	Payment proceeds with no change to the customer experience.	99.63%
STEP-UP	Scam-specific warning and explicit re-confirmation. Also triggered by behaviour never seen in genuine traffic.	0.32%
HOLD	Payment paused and customer informed. Released after verification by the customer or bank.	0.05%

Example reasons attached to alerts: *first payment to this receiver*; *customer on a live call while paying*; *receiver paid by 7 different people today*; *device not previously seen on this account*.

CUSTOMER MESSAGE · HOLD · ₹95,000

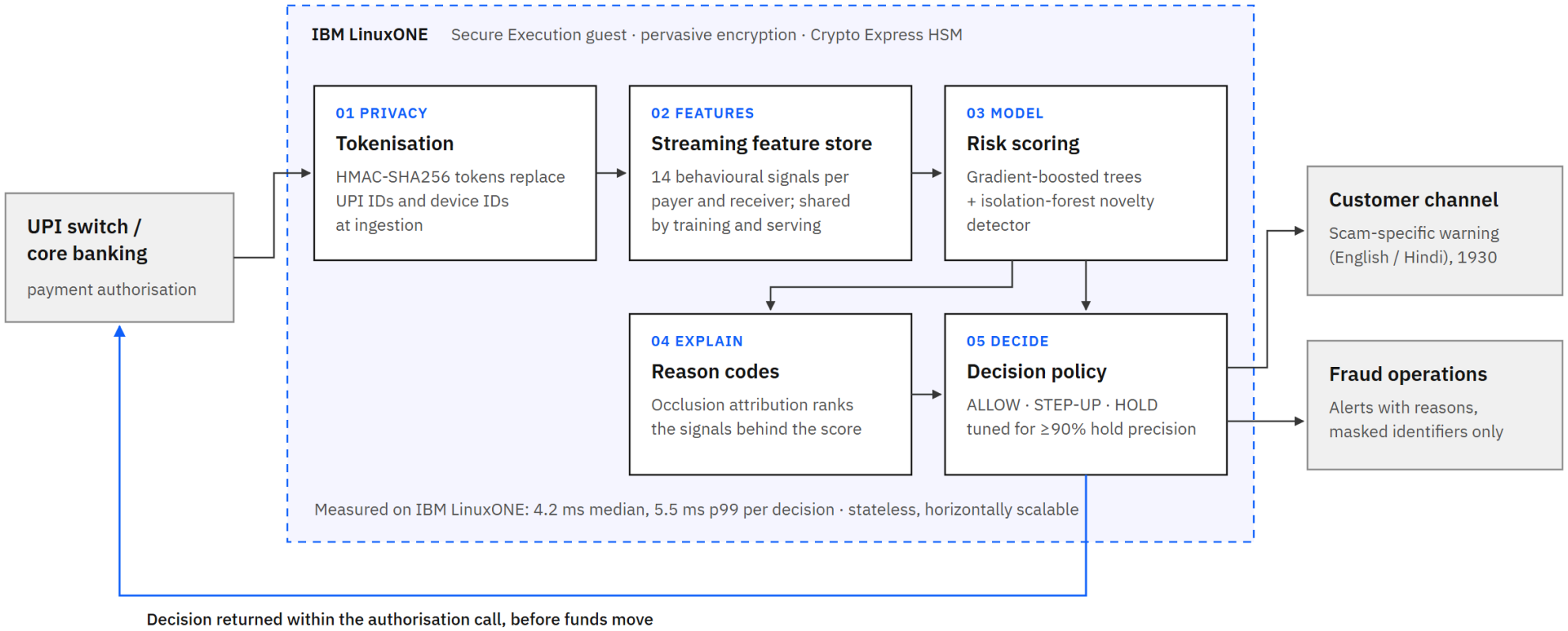
No police, CBI, customs or bank officer will ever ask you to transfer money on a call. Hang up and dial 1930 (Cyber Crime Helpline).

कोई भी पुलिस, CBI, कस्टम या बैंक अधिकारी कॉल पर पैसे ट्रांसफर करने को नहीं कहता। कॉल काटें और 1930 पर संपर्क करें।

The message is selected from the signals that drove the decision and shown in English and Hindi.

ARCHITECTURE

Scoring runs on LinuxONE inside the authorisation path. Identifiers are tokenised before any feature or model sees them.



METHOD

Behavioural features computed per payment feed two complementary models, evaluated strictly on future data.

Features

14 signals: amount relative to the payer's norm, first-time receiver, device and location, live-call flag, COLLECT type, payment bursts, 24-hour spend, receiver fan-in and account age. A receiver, device or city becomes trusted only after 24 hours.

Models

Gradient-boosted trees learn known fraud patterns. An isolation forest trained only on genuine traffic flags unfamiliar behaviour for step-up, giving coverage beyond the patterns in the training labels.

Explanation

Occlusion attribution: each signal is reset to a typical genuine value and the drop in risk is measured. This runs in one batched model call, and the top signals become reason codes.

Evaluation

Time-based split: train on days 1–17, set thresholds on days 18–22, report on days 23–30. The same feature code is used in training replay and live serving, so there is no train/serve skew.

Data. 139,755 simulated UPI payments over 30 days (4,000 customers, 600 merchants, 80 mule accounts). It includes four fraud patterns from public advisories and deliberately includes hard cases: large payments to new legitimate receivers, genuine payments made during calls, new merchants, travel, device changes, and long-established accounts used as mules. Real UPI data is not publicly available.

RESULTS · HELD-OUT FINAL WEEK, 32,854 PAYMENTS, 437 FRAUDS

ZeroLag intercepts 96% of scams and 98% of fraud value while affecting fewer than 4 in 1,000 genuine payments.

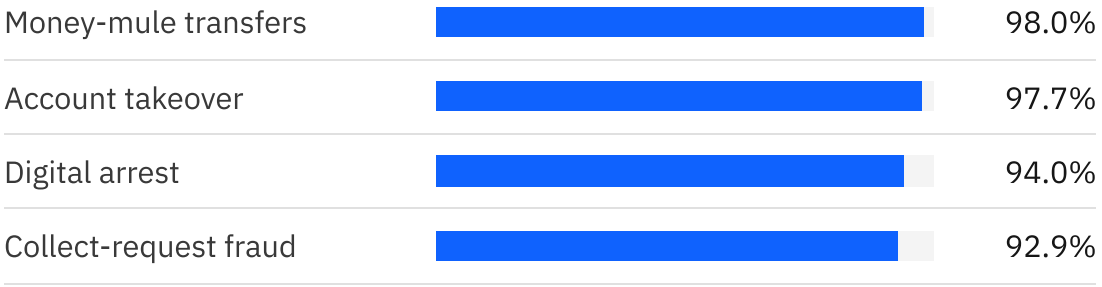
95.9%
Frauds warned or held (recall)

98.3%
Fraud value intercepted

96.2%
Precision of HOLD decisions

4.2 / 5.5 ms
p50 / p99 decision latency on IBM LinuxONE, including explanation

DETECTION RATE BY FRAUD TYPE



OPERATING METRIC	VALUE
Genuine payments with any friction	0.37%
Genuine payments held	0.05%
PR-AUC	0.972
ROC-AUC	0.999

Results are on simulated data. Production use requires validation on bank data, drift monitoring and periodic re-calibration. All figures from a training and evaluation run on IBM LinuxONE (s390x, machine type 9175); latency single-threaded.

MEASURED ON IBM LINUXONE

On LinuxONE, ZeroLag decides in 4.2 ms, three times faster than on commodity hardware and with a five times tighter tail.

SAME CODE, SAME DATA	LAPTOP (X86)	IBM LINUXONE
Decision latency, median	13 ms	**4.2 ms**
Decision latency, 99th percentile	27 ms	**5.5 ms**
Feature replay, 139,755 payments	11.8 s	**4.0 s**
Full training and evaluation run	57 s	**17 s**
Live dashboard, average per decision	11–15 ms	**5–6 ms**

A payment authorisation call has a fixed time budget, so the predictable tail matters more than the average. The guest also exposes the **NNPA** facility. The Telum II on-chip AI accelerator is available for the next step: serving the model on it.

SYSTEM EVIDENCE (LSCPU, UNAME)

```
Architecture:   s390x
OS:             RHEL 9.8 (kernel el9_8.s390x)
Vendor ID:      IBM/S390
Machine type:   9175
CPU MHz:        5508
CPU(s):         3
Hypervisor:     z/VM 7.3.0
Flags:          ... vxe2 vxp2 nnpa sie
Python:         3.12.15
scikit-learn:   1.9.1, built on-host for z15
```

Full output: docs/linuxone_proof.txt and artifacts/metrics_linuxone.json in the repository. Laptop: Windows 11, x86-64, same single-thread setting.

RESPONSIBLE AI AND PRIVACY

Protection is consistent across age groups, and no step of the pipeline handles a raw customer identifier.

CONTROL	IMPLEMENTATION
Identity protection	UPI and device IDs are replaced by keyed HMAC-SHA256 tokens at ingestion. Analysts see masked IDs only.
No protected attributes	Age and other demographics are not model inputs. Age is used only for the outcome audit.
Explainability	Every STEP-UP and HOLD carries ranked reason codes.
Human oversight	HOLD pauses a payment and does not reverse it. Release follows verification.

OUTCOME AUDIT BY PAYER AGE GROUP

AGE GROUP	GENUINE PAYMENTS FLAGGED	FRAUDS DETECTED
18–35	0.31%	92.7%
36–55	0.42%	96.7%
56 and over	0.36%	97.4%

Older customers, who are disproportionately targeted by digital-arrest fraud, receive the same level of protection without a higher false-alarm rate.

PROTOTYPE

The working prototype scores a live transaction stream and exposes a REST scoring API with an analyst console.



Analyst console: live KPIs, held payments with reason codes and customer messages, interactive scenario testing through POST /api/score, and a masked transaction stream.

WHY IBM Z AND LINUXONE

Payment systems already run on IBM Z. Scoring there removes a network hop, a second copy of sensitive data and an external point of failure.

REQUIREMENT	IBM Z CAPABILITY	ZEROLAG TODAY → PRODUCTION PATH
In-transaction inference at scale	Telum on-chip AI accelerator; IBM Snap ML and Z Deep Learning Compiler; Machine Learning for z/OS for CICS/IMS scoring	Runs on LinuxONE today at 4.2 ms p50; the guest exposes the NNPA facility → tree model served on the Telum II accelerator
Key protection	Crypto Express HSM (CCA / EP11); pervasive encryption	HMAC tokenisation key from configuration → key held and used inside the HSM
Confidential processing	IBM Secure Execution for Linux; Hyper Protect Virtual Servers	Stateless container → deployed as a Secure Execution guest
Long-term cryptographic safety	Quantum-safe algorithms (ML-KEM, ML-DSA) on IBM Z	Standard TLS → quantum-safe TLS and key wrapping, no model change
Availability	IBM Z reliability and vertical scale	Stateless scoring replicated across cores and LPARs

IMPACT AND ROADMAP

ZeroLag protects the customers most exposed to payment fraud and gives banks an auditable, low-friction control.

Beneficiaries

- Elderly and first-time digital payment users
- Banks and payment service providers: fewer losses, disputes and refunds
- Law enforcement: earlier identification of mule accounts

Next three months

- Serve the model on the Telum accelerator
- Move tokenisation keys into Crypto Express
- Deploy as a Secure Execution guest
- Pilot on anonymised bank data

Beyond

- Privacy-preserving mule-account signals shared across banks
- Additional Indian languages for customer messages
- Continuous drift and fairness monitoring